Momentum-Based Multi-Strategy Trading System

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Abstract

This document presents a tactical allocation system built from four complementary building blocks: regional equity momentum, gold versus S&P 500 rotation, growth-and-inflation sector timing, and a bond allocation with an interest-rate-regime filter. The system combines equity trend following, macro-sensitive sector rotation, defensive gold exposure, and fixed-income exposure into a single monthly rebalanced portfolio. The strategy aims, approximately and over long investment horizons, to achieve an annual gain of about 20%, a Sharpe ratio close to 2, and a maximum drawdown around 10%. These targets are design objectives rather than guaranteed outcomes.

1 Introduction

This document provides a standalone specification of a tactical multi-strategy investment system. The portfolio is organized as eight independent strategy components. Each component produces a monthly equity curve, and the final portfolio combines their monthly return streams using fixed capital weights. Signals are computed from adjusted closing prices and reviewed at month-end.

The system is designed to diversify across three dimensions: geographical equity universes, macro regimes, and asset classes. Regional momentum components provide stock-level trend exposure across several developed equity markets. The gold rotation component introduces a defensive relative-strength switch between gold and the S&P 500. The sector strategy captures macro-sensitive sector leadership across growth and inflation regimes. The bond component adds fixed-income exposure through an equal-weighted risk-on basket, with an additional yield-curve filter that switches the component into SHY when the 10-year minus 3-month Treasury spread is not positive.

The strategy aims, approximately and over long investment horizons, to achieve an annual gain of about 20%, a Sharpe ratio close to 2, and a maximum drawdown around 10%. These targets are strategic objectives for the allocation process, not guaranteed outcomes. Realized performance can differ materially depending on market conditions, transaction costs, taxes, slippage, data quality, and the exact investment universe used.

2 Strategy Overview

The strategy uses eight return streams: five regional equity momentum components plus gold rotation, sector timing, and bond allocation. Capital allocation is fixed at the component level and normalized before calculating the combined equity curve.

Table 1: Strategy components and target portfolio weights.

Component	Description	Portfolio weight
EUR	European stock momentum universe, built from STOXX Europe 600 components.	10%
USD	U.S. stock momentum universe, built from S&P 500 components.	10%
AUD	Australian stock momentum universe, built from S&P/ASX 200 components.	10%
GBX	U.K. stock momentum universe, built from FTSE 350 components.	10%
CAD	Canadian stock momentum universe, built from S&P/TSX Composite components.	10%
GOLD	Gold versus S&P 500 rotation.	10%
SECT	Growth-and-inflation sector timing.	10%
BOND	Equal-weighted bond ETF basket with yield-curve switch to SHY.	30%

3 Data Acquisition and Preprocessing

The regional equity universes are created from locally saved TradingView component pages. Each component list is filtered by listing currency and converted to Yahoo Finance ticker symbols using the appropriate exchange suffix. Historical prices are downloaded as adjusted closing prices. Adjusted prices are used so that dividends and corporate actions are reflected consistently in the price series.

All trading rules are evaluated at monthly frequency. Daily prices are converted to month-end observations for signal generation and performance calculation. Abnormal price jumps are filtered by detecting very large close-to-close ratios; price histories preceding such discontinuities are excluded from the active sample.

4 Regional Equity Momentum Strategy

The regional equity momentum strategy is applied independently to each currency universe. It ranks individual stocks by long-term momentum, retains existing positions while short-term momentum remains positive, and uses a market-breadth condition to reduce exposure during broad bearish phases.

4.1 Investment Universes and Source Files

The regional momentum strategy is applied to five equity universes, each corresponding to the component list of a major equity index. The component lists are sourced from TradingView pages. Operationally, each TradingView component page is opened in Google Chrome, the full list is loaded using the page's *Load more* button, and the fully loaded page is saved locally in MHTML format.

Table 2: Regional equity universes used by the momentum strategy.

Currency	Investment universe	TradingView page	Local MHTML source file
USD	S&P 500 Index components	SPX components	`S&P 500 Index Components - SP_SPX Stocks - TradingView.mhtml`
EUR	STOXX Europe 600 components	TVC-SXXP components	`STOXX 600 Components - TVC_SXXP Stocks - TradingView.mhtml`
GBX	FTSE 350 Index components	FTSE-NMX components	`FTSE 350 Index Components - FTSE_NMX Stocks - TradingView.mhtml`
CAD	S&P/TSX Composite components	TSX-TSX components	`S&P_TSX Composite Index Components - TSX_TSX Stocks - TradingView.mhtml`
AUD	S&P/ASX 200 components	ASX-XJO components	`S&P_ASX 200 Index Components - ASX_XJO Stocks - TradingView.mhtml`

Because index membership evolves over time and TradingView pages are dynamic, the MHTML files should be refreshed periodically. The backtest uses the locally saved files as the investable universe input.

4.2 Momentum Indicators

For each stock i and date t, the normalized momentum signal over horizon d is defined as the distance between the adjusted close and its rolling moving average, divided by the adjusted close:

$$M_{d,i}(t) = \frac{P_i(t) - \mathrm{MA}_{d,i}(t)}{P_i(t)}, \tag{1}$$

where $P_i(t)$ is the adjusted close and $\mathrm{MA}_{d,i}(t)$ is the rolling average over $d+1$ daily observations.

Two horizons are used:

- M_{240}, a long-term momentum indicator used for stock ranking and new entries;
- M_{90}, a short-term continuation indicator used to retain or exit existing positions.

4.3 Market Breadth Condition

The strategy also measures the breadth of positive long-term momentum across the full regional universe. The breadth oscillator is defined as

$$B(t) = \frac{N_{\mathrm{pos}}(t) - N_{\mathrm{neg}}(t)}{N_{\mathrm{pos}}(t) + N_{\mathrm{neg}}(t)}, \tag{2}$$

where $N_{\mathrm{pos}}(t)$ is the number of stocks with positive M_{240} and $N_{\mathrm{neg}}(t)$ is the number of stocks with negative M_{240}.

The oscillator $B(t)$ lies between -1 and $+1$. A positive and sufficiently high value indicates broad participation in the trend, while a low or negative value indicates weak market breadth.

4.4 Portfolio Construction

At each month-end review date, the regional strategy applies the following rules:

1. Carry forward existing positions whose M_{90} signal remains positive.

2. Sort all stocks by M_{240} in descending order.
3. Add new positions, in rank order, only when $M_{240} > 0$, $M_{90} > 0$, $B(t) > 0.1$, the adjusted close is non-zero, and the stock is not already held.
4. Stop adding positions once ten stocks are held.
5. Assign each selected stock a fixed 10% weight within the regional portfolio. If fewer than ten stocks qualify, the unused capital remains in cash.

Positions selected at the close of month t are applied to returns over the following month. Each regional component therefore behaves as a concentrated but diversified momentum portfolio with an explicit cash buffer during weak market regimes.

5 Gold Versus S&P 500 Rotation Strategy

The gold component is a monthly relative-strength rotation between the S&P 500 ETF SPY and the gold ETF GLD. It is inspired by the gold and S&P 500 rotation concept described by Quantified Strategies [1].

Universe: SPY and GLD, or their European-listed equivalents PSPH.PA and SGLD.AS, respectively.

5.1 Signal Construction

The strategy compares gold to equities through the ratio

$$R_{\text{gold}}(t) = \frac{\text{GLD}(t)}{\text{SPY}(t)}. \tag{3}$$

The trend reference is the 36-month simple moving average of the ratio:

$$T_{\text{gold}}(t) = \text{SMA}_{36}\left(R_{\text{gold}}(t)\right). \tag{4}$$

5.2 Allocation Rule

The allocation rule is binary:

$$w_{\text{GLD}}(t) = \begin{cases} 1, & R_{\text{gold}}(t) > T_{\text{gold}}(t), \\ 0, & R_{\text{gold}}(t) \le T_{\text{gold}}(t), \end{cases} \qquad w_{\text{SPY}}(t) = 1 - w_{\text{GLD}}(t). \tag{5}$$

The component therefore allocates fully to gold when gold exhibits stronger relative strength than equities, and otherwise allocates fully to the S&P 500.

6 Growth and Inflation Sector Timing Strategy

The sector timing component is a regime-based equity sector rotation model inspired by David Varadi's Growth and Inflation Sector Timing Model [2]. It combines an inflation proxy with a growth proxy and maps the resulting regime to a target equity sector.

Universe: SPY, XLB, XLE, XLF, XLI, XLK, XLP, XLU, and XLV.

6.1 Inflation Proxy

Expected inflation is proxied by the relative strength of a cyclical, inflation-sensitive basket against a defensive basket:

$$I(t) = \frac{0.50\,\mathrm{XLE}(t) + 0.17\,\mathrm{XLI}(t) + 0.17\,\mathrm{XLF}(t) + 0.16\,\mathrm{XLB}(t)}{0.34\,\mathrm{XLP}(t) + 0.33\,\mathrm{XLU}(t) + 0.33\,\mathrm{XLV}(t)}. \tag{6}$$

For each lookback horizon $L \in \{12, 24, 36, 48\}$ months, expected inflation is classified as high when $I(t)$ is above its rolling median over L months.

6.2 Growth Proxy and Regime Mapping

Expected growth is proxied by SPY. For each lookback horizon L, the growth condition is classified as bullish when SPY is above its rolling median over the same horizon. The sector target is then determined by the two-dimensional regime map in Table 3.

Table 3: Growth-and-inflation regime map and corresponding sector ETFs.

Inflation condition	Growth condition	Sector selected	ETF	ETF Europe
High inflation	Bullish growth	Energy	XLE	QDVF.DE
High inflation	Bearish growth	Health Care	XLV	QDVG.DE
Low inflation	Bullish growth	Technology	XLK	ZPDT.DE
Low inflation	Bearish growth	Consumer Staples	XLP	2B7D.DE

Because four lookback horizons are combined, the final sector exposure can be split across several ETFs when the horizons disagree. Each horizon contributes 25% of the sector component.

7 Bond Yield-Curve Allocation Strategy

The bond component is no longer a cross-sectional momentum ranking strategy. In the updated implementation, it is a regime-filtered equal-weight allocation applied to a diversified risk-on fixed-income and convertible-bond ETF basket. The defensive asset SHY is deliberately excluded from the normal risk-on basket; it is used only when the yield-curve filter indicates an unfavorable regime.

Universe: the strategy uses the U.S. ETFs and European-listed equivalents shown in Table 4. In the U.S. implementation, IEF, LQD, HYG, TIP, and CWB form the equal-weighted investable basket when the yield curve is positively sloped. SHY is the defensive allocation used when the curve is flat or inverted.

Table 4: Bond ETF universe and European-listed equivalents.

ETF US	ETF Europe	Role in the updated strategy
SHY	IBTE.L	Defensive Treasury allocation used only when the yield-curve filter is inactive.
IEF	IBB1.DE	Risk-on bond basket, 20% weight when the curve is positive.
LQD	VDCE.DE	Risk-on bond basket, 20% weight when the curve is positive.
HYG	UEEF.DE	Risk-on bond basket, 20% weight when the curve is positive.
TIP	IBC5.DE	Risk-on bond basket, 20% weight when the curve is positive.
CWB	ZPRC.DE	Risk-on bond basket, 20% weight when the curve is positive.

7.1 Yield-Curve Filter

The macro risk filter is based on the FRED series T10Y3M, defined as the 10-year Treasury constant maturity rate minus the 3-month Treasury constant maturity rate. Let

$$Y(t) = y_{10\mathrm{Y}}(t) - y_{3\mathrm{M}}(t), \tag{7}$$

where $y_{10\mathrm{Y}}(t)$ and $y_{3\mathrm{M}}(t)$ are the corresponding month-end Treasury yields. The risk-on bond basket is active only when the curve is positively sloped:

$$F_{\mathrm{bond}}(t) = \begin{cases} 1, & Y(t) > 0, \\ 0, & Y(t) \leq 0. \end{cases} \tag{8}$$

When $F_{\mathrm{bond}}(t) = 1$, the component is invested equally across the five risk-on ETFs. When $F_{\mathrm{bond}}(t) = 0$, the component is invested fully in SHY.

7.2 Allocation Rule

At each month-end:

1. compute the month-end value of the T10Y3M yield-curve spread;
2. if $Y(t) > 0$, allocate 20% to each of IEF, LQD, HYG, TIP, and CWB;
3. if $Y(t) \leq 0$, allocate 100% to SHY;
4. apply the resulting weights to the following month's returns.

Equivalently, with the ordered risk-on basket

$$\mathcal{B} = \{\mathrm{IEF}, \mathrm{LQD}, \mathrm{HYG}, \mathrm{TIP}, \mathrm{CWB}\}, \tag{9}$$

the final weights are

$$w_i(t) = \begin{cases} 1/5, & i \in \mathcal{B} \text{ and } Y(t) > 0, \\ 1, & i = \mathrm{SHY} \text{ and } Y(t) \leq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

This design removes the previous cross-sectional ranking step and makes the bond component a transparent regime switch: broad equal-weighted fixed-income exposure in normal yield-curve regimes, and short-duration Treasury exposure during curve inversions or flat-curve periods.

8 Composite Portfolio Construction

Each component produces a standalone monthly equity curve. The combined portfolio return is calculated as the weighted sum of these component returns using the fixed coefficients shown in Table 1. The final equity curve is obtained by compounding the resulting monthly portfolio returns:

$$E_{\text{combo}}(t) = E_{\text{combo}}(t-1)\left[1 + \sum_{j=1}^{8} \alpha_j r_j(t)\right], \tag{11}$$

where α_j is the fixed capital weight of component j and $r_j(t)$ is its monthly return.

This construction separates signal generation from capital allocation. Stock selection takes place inside the regional portfolios, ETF rotation takes place inside the gold, sector, and bond components, and the final portfolio combines the resulting return streams through fixed strategic weights.

9 Strategy Summary

Table 5: Summary of the four strategy families.

Strategy	Main signal	Rebalance	Risk control / cash rule
Regional Momentum	M_{240} ranking with M_{90} continuation filter and breadth $B(t)$.	Monthly	New entries require $B(t) > 0.1$; unfilled slots remain in cash.
Gold Rotation	GLD/SPY ratio versus its 36-month moving average.	Monthly	Fully invested in either GLD or SPY.
Sector Timing	Inflation proxy and SPY growth proxy versus rolling medians over 12, 24, 36, and 48 months.	Monthly	Exposure is split across four regime horizons.
Bond Yield-Curve Allocation	Equal-weighted basket of IEF, LQD, HYG, TIP, and CWB, gated by the 10Y–3M Treasury spread.	Monthly	20% in each risk-on ETF when T10Y3M is positive; 100% in SHY when T10Y3M is not positive.

10 Backtest Procedure and Results

The reference backtest starts on 1 January 2004. It evaluates each component monthly, records recent monthly positions for reporting, plots the individual equity curves, and plots the combined equity curve. The reported statistics are annualized mean return, annualized volatility, annualized Sharpe ratio, and maximum drawdown.

The backtest does not model transaction costs, taxes, bid-ask spreads, execution constraints, borrow costs, survivorship effects beyond the chosen universe files, or currency conversion frictions. These implementation factors should be considered before live deployment.

10.1 Individual Strategy Backtests

The eight standalone return streams are evaluated independently before being combined: EUR, USD, AUD, GBX, and CAD regional equity momentum strategies, plus the Gold, Sector, and Bond components. Figure 1 shows the individual equity curves on a logarithmic scale.

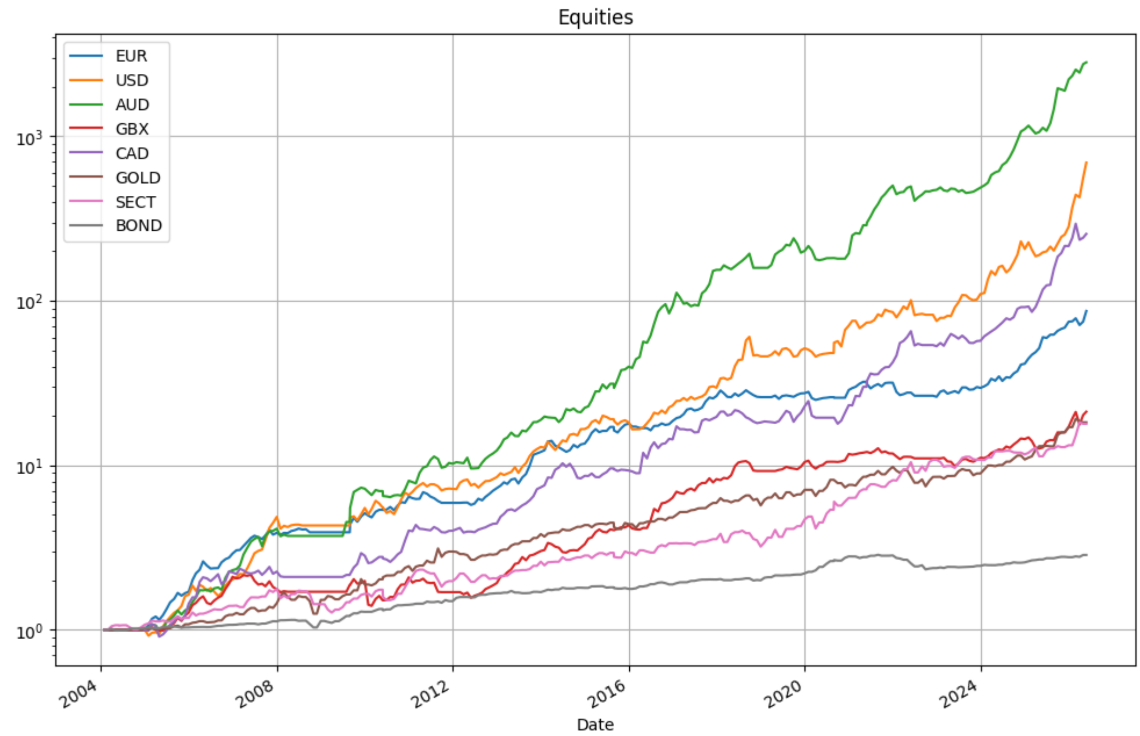

Figure 1: Individual strategy equity curves, log scale.

10.2 Combined Portfolio Backtest

The combined portfolio is obtained as the weighted sum of the monthly returns of the eight components, using the strategic weights defined in Table 1. Figure 2 shows the resulting combined equity curve on a logarithmic scale. The bond component uses the updated equal-weighted bond basket and the SHY switch described above.

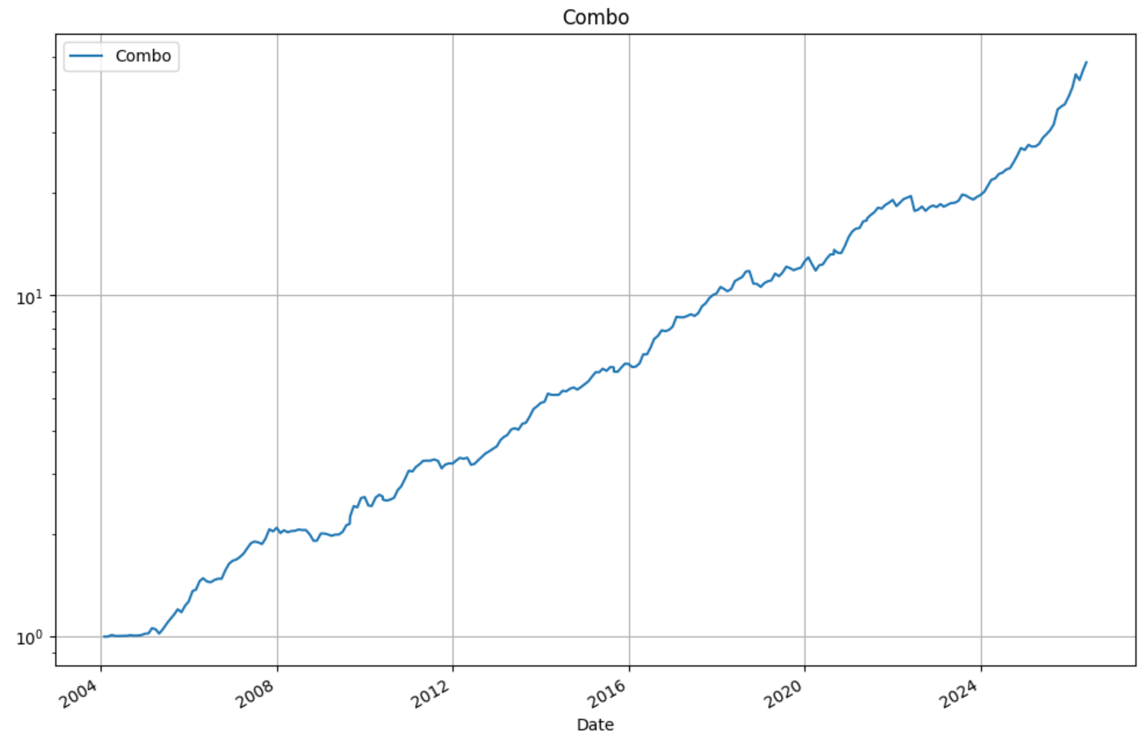

Figure 2: Combined portfolio equity curve with the strategic weights and the updated bond yield-curve switch, log scale.

Table 6: Combined portfolio performance statistics.

Metric	Value
Annualized mean return	17.51%
Annualized volatility	9.24%
Annualized Sharpe ratio	1.90
Maximum drawdown	10.15%

The combined profile remains broadly consistent with the target range stated in the introduction: an annualized return close to 20%, a Sharpe ratio close to 2, and a maximum drawdown close to 10%. The updated notebook reports an annualized mean return of 17.51%, annualized volatility of 9.24%, a Sharpe ratio of 1.90, and a maximum drawdown of 10.15%.

References

[1] Quantified Strategies, "Rotation Strategy in S&P 500 and Gold (SPY & GLD)." https://www.quantifiedstrategies.com/rotation-strategy-in-gold-and-sp500/

[2] David Varadi, "The Growth and Inflation Sector Timing Model," CSS Analytics, 20 March 2025. https://cssanalytics.wordpress.com/2025/03/20/the-growth-and-inflation-sector-timing-model/

[3] Cambria Investments, "T-Bills and Chill (Most of the Time)," February 2025. https://www.cambriainvestments.com/wp-content/uploads/2025/02/20250210.TBillsandChillMostoftheTime.Approved.pdf

[4] Federal Reserve Bank of St. Louis, FRED series T10Y3M. https://fred.stlouisfed.org/series/T10Y3M